AI TOOLS FOR RESEARCH

AI-assisted coding, vibe coding and evidence-based experimentation

THE CHALLENGE

Your first paper mapped the research landscape. Your second paper must ask a focused question, build or reproduce a working solution, and investigate it through real, reproducible experiments.

You have already completed a literature review on a research topic. Use that review as your evidence base, not as text to recycle. Verify the original studies, select one tractable problem, establish a baseline, use AI tools to implement and debug code, evaluate a meaningful research contribution, and write a second paper supported by measured results.

Research journey REVIEW → VERIFIED GAP → QUESTION → BASELINE → VALIDATION → EXPERIMENT → EVIDENCE → PAPER → REPRODUCE

A successful project includes an evidence-based research question, actual executable code, a verified baseline, a fair experimental comparison, a meaningful analysis, a complete paper and a documented repository. A carefully analyzed negative result can be as valuable as an improvement.

1 | Learning outcomes

- Connect your implementation study explicitly to verified findings from your previous literature review.
- Formulate research questions, hypotheses (when applicable) and measurable success criteria.
- Use AI-assisted coding and vibe coding to specify, generate, read, run, test, debug and improve code.
- Reproduce or validate a trusted baseline before claiming benefits from any modification.
- Design fair comparisons, prevent data leakage and analyze real results critically.
- Write an evidence-grounded implementation paper with LaTeX/Overleaf and reproducible figures.
- Document the human–AI collaboration and explain technical decisions without relying on AI output as authority.

2 | The research standard

This is not a task to ask AI for a complete paper. It is a task to conduct an experimental study with AI as a supervised research assistant.

Continuity. Remain in the research area of your completed review. Narrow the scope if required.

Execution. Produce working and tested code. Screenshots, mock-ups or application interfaces alone are insufficient.

Evidence. All reported numbers, tables, plots and conclusions must come from actual experiments or explicitly attributed sources.

Comparison. Include two appropriate baselines/comparators where feasible; justify an equivalent rigorous design when not.

Depth. Add at least one deeper analysis: ablation, error or failure cases, robustness, sensitivity or resource efficiency.

Reproducibility. Record the data protocol, environment, parameters, random seeds and commands needed to regenerate key results.

Ownership. You must understand the submitted code, trace its origin, validate AI suggestions and explain important decisions.

IMPORTANT DISTINCTION

You are not required to invent an entirely new algorithm. You are required to conduct a meaningful, technically correct experiment that answers a research question. Course-level research quality and publication-level novelty are not the same.

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3 | Approved pathways to a contribution

Pathway	What you do	Evidence needed
A. Reproduction + extension	Reproduce a published method, then study a justified new condition or component.	Validated baseline; extension comparison; explanation of discrepancies.
B. Controlled comparative study	Evaluate relevant methods under a common, carefully chosen protocol that answers a specific question.	Fair controls; consistent metrics; informative failure or trade-off analysis.
C. Targeted modification	Change one technically justified component of a baseline.	Baseline preserved; isolated change; ablation or controlled test.
D. Hybrid/prototype + evaluation	Combine methods or build a prototype to investigate a computational hypothesis.	Motivation for design; measurable research outcomes; baselines and rigorous evaluation.

Weak submission: "I used AI to build a model and achieved 94% accuracy." Strong submission: "I investigated a literature-supported question using a validated baseline, controlled comparisons, documented code, and an analysis explaining when and why the method succeeds or fails."

4 | Milestones and approval gates

Gate	What students must submit	Ready to proceed when…
G1 Research brief	1-page gap/problem brief; original supporting papers; question; pathway; compute and data feasibility.	The question is narrow, literature-supported and experimentally testable.
G2 Baseline verification	Accessible dataset; environment test; runnable baseline; measured output; validation report.	Baseline behavior and protocol have been checked, not merely executed.
G3 Experimental evidence	Frozen comparison protocol; outputs/logs; result table and plots; one deeper analysis.	Experimental claims are traceable to genuine runs and evaluated fairly.
G4 Paper and reproducibility	Paper PDF + LaTeX; GitHub repository; AI audit; 5-minute demonstration.	Methods, outputs and provenance can be inspected and defended.

Suggested pacing: four weekly milestones (one gate per week), or as announced by the instructor. Official deadlines, group policy and submission location will be communicated through Google Classroom.

5 | First laboratory: the 2-hour launch

Time	Activity	Output
0–20 min	Revisit your review and annotate original paper evidence.	Candidate gaps and paper references.
20–45 min	Use Prompts 01–03; select one experiment path.	Literature-to-experiment map.
45–70 min	Verify novelty, dataset access and feasibility; frame your question.	Draft research brief and feasibility record.
70–105 min	Identify a baseline; configure a notebook and run a small environment/data sanity check.	Code/notebook with actual output.
105–120 min	Record decisions and submit the launch checkpoint.	One-page plan, notebook link and AI log entry.

DAY-ONE RULE

The first lab is successful when you have a feasible question, an identifiable baseline and a verified first execution step. You do not need to finish the entire research implementation in two hours.

6 | AI Prompt Studio: copy, adapt, verify

Use the 18 prompts in sequence. Replace bracketed text with your own files, evidence, code and results. Prompts are working scaffolds, not instructions to surrender judgment. Save the useful outputs in your research repository, not only in chat history.

At every stage, maintain an evidence ledger with three labels: VERIFIED (checked from original source or actual execution), INFERRED (plausible interpretation requiring testing) and UNKNOWN (missing evidence). If your AI tool lacks browsing or file access, supply sources manually. Never accept a citation or repository solely because it looks plausible.

A worked example: what the transition looks like

Illustration only (not a pre-verified literature claim): suppose your review discusses image classification under changing lighting conditions. After checking the original papers, you decide to study whether classification robustness degrades under controlled illumination shifts.

From the review	To the executable study
Observed limitation	Sensitivity under changing illumination, supported by an original study you verify.
Research question	How do two classifiers perform under a consistent brightness shift?
Baseline and test	Run verified models on the same data partition; apply the shift only for a prespecified robustness test.

From the review	To the executable study
Evidence and contribution	Report macro-F1, per-class errors and efficiency; explain observed robustness differences without claiming novelty by default.

Use this example for the pattern of reasoning, not as a topic to copy. Your own question must arise from your own verified review.

PHASE I — FROM REVIEW TO RESEARCH QUESTION

PROMPT 01 | Set up an accountable research workspace

I am a postgraduate student converting my completed literature review into an implementation-based research paper. My topic is [TOPIC]. Act as a critical research mentor and technical pair programmer. Guide me in stages: evidence-based gap identification, literature-to-experiment mapping, feasibility, research questions, verified baseline, coding, experimental evaluation, manuscript and reproducibility. Mark statements as VERIFIED, INFERRED or UNKNOWN, and explicitly request source files or outputs when evidence is missing. Never invent papers, datasets, repository links, code execution, experiments or metrics. Help me create a portable research_decisions.md log containing decisions, reasons, source/evidence and next actions. Start by asking me to attach my existing review and summarize its scope without assuming unseen content.

SAVE: Workspace and decision log; review scope.

PROMPT 02 | Extract evidence-based research gaps

Analyze my attached literature review [ATTACH PAPER]. Identify central methods, datasets, measured findings, author-reported limitations and unresolved questions. Propose five computationally testable research gaps. For each, cite the specific passage or reference in my review; clearly distinguish a limitation reported in the original paper from your own inference; state the experimental question it suggests and what evidence would test it. Flag any claims needing verification against the original publications. Rank three gaps by research significance, available resources and implementation feasibility. Do not claim novelty without a separate literature search.

SAVE: Gap matrix: source evidence → candidate question → possible experiment.

PROMPT 03 | Map the literature to an executable experiment

My literature review [ATTACH REVIEW] identified these candidate gaps: [GAPS]. Identify three published methods from my review that could realistically be implemented or reproduced. For each method, map its original paper to the algorithm, dataset, code or model access, evaluation metric, known limitations and computational requirement. Distinguish verified access to code/weights from unverified suggestions. Compare three possible experimental studies: (A) reproduction plus extension, (B) controlled comparison, and (C) targeted modification. For each study state the smallest working baseline, the testable question, expected resource burden and evidence required. Recommend one minimum viable research experiment for [HARDWARE] and [TIME AVAILABLE]. Provide a literature-to-experiment mapping table and list what I must personally verify before coding. Do not propose an elaborate new architecture without a scientific rationale.

SAVE: Mandatory bridge: original papers → selected method → executable baseline → planned test.

PROMPT 04 | Check feasibility, novelty and resource access

My candidate study is [IDEA]. Develop a feasibility and novelty-verification plan. Provide scholarly search strings and specific categories of neighboring work to inspect, without inventing titles, DOIs or URLs. Create a matrix with columns: published method, VERIFIED contribution (with source), limitation, proposed difference and evidence still missing. Check whether the dataset, code, model weights, license and benchmark protocol are accessible; flag all unresolved availability claims. Estimate computational and time constraints for [HARDWARE] and [WEEKS]. Narrow the study to a viable

experimental question and recommend a fallback if the preferred repository, weights or dataset are unavailable.

SAVE: Novelty/feasibility matrix and go/no-go decision.

PROMPT 05 | Frame research questions and success criteria

Here is my verified gap [GAP], selected study [IDEA] and source evidence [SOURCES]. Write one precise research problem, two or three testable research questions, and hypotheses only where appropriate. For each question identify the input data, comparator, independent/dependent variables when relevant, evaluation metric, meaningful success threshold or decision rule, and an observation that would contradict my expectation. Separate the contribution I intend to investigate from a contribution already demonstrated. Flag overbroad questions and provide a one-page research brief that can be reviewed before implementation.

SAVE: One-page problem brief with questions and measurable outcomes.

PHASE II — EXPERIMENT PLAN

PROMPT 06 | Verify datasets and comparators

My question is [RESEARCH QUESTION] in [TASK/DOMAIN]. Recommend a primary dataset and two meaningful baseline methods or comparable controls, where feasible. For each candidate, identify the official source to verify, file structure, benchmark protocol, licensing/access terms, approximate compute requirement and common pitfalls. Explain an evaluation design appropriate to this domain: train/validation/test partitions when relevant, or a justified alternative for systems/HCI/security studies. Specify likely leakage risks such as patient-, subject-, document-, company- or time-level overlap as applicable. Do not present unverified access claims as facts. End by recommending the simplest credible baseline and a short independent-verification checklist.

SAVE: Dataset and baseline validation sheet.

PROMPT 07 | Freeze an experiment blueprint before viewing results

Design a reproducible protocol for [QUESTION] on [DATASET] comparing [BASELINES] with [PROPOSED STUDY]. Include precise input/output definitions, modules, preprocessing, Mermaid architecture diagram, data partitions, controls, training or processing settings, evaluation metrics (with whether higher/lower is better), random seed plan, computational budget and expected table/figure formats. Include at least one deep analysis: ablation, error/failure, robustness, efficiency or sensitivity. Define threats to validity and a pre-experiment checklist to freeze the primary metric, test split and comparison protocol before viewing results. Explain how legitimate protocol changes should be logged later. Never fabricate anticipated outcomes.

SAVE: Frozen evaluation protocol, flowchart and pre-experiment checklist.

PHASE III — AI CODING, VIBE CODING AND BASELINE VALIDATION

PROMPT 08 | Configure and test the coding environment

Act as my research software engineer. Set up [GOOGLE COLAB/LOCAL/INSTITUTIONAL GPU] for [METHOD] using [LIBRARIES]. Propose dependency versions and show how to check their compatibility instead of claiming untested compatibility. Provide installation commands, GPU/system diagnostics, dataset access procedures, project directories, random-seed controls and a clean repository structure (README, src, notebooks, configs, scripts, results, paper, requirements). Write a small executable environment test, state its expected output type and list failure diagnostics. Stop after this test and ask me to share the actual console output before proceeding.

SAVE: Environment verification log and structured project repository.

PROMPT 09 | Vibe-code one baseline module at a time

> You are my AI pair programmer. I am reproducing [PUBLISHED BASELINE] for [DATASET] to produce [OUTPUT]. Work in short cycles: (1) specify the module and assumptions, (2) generate minimal executable code, (3) explain the logic and input/output shapes, (4) supply a unit/sanity test, and (5) wait for my genuine execution output before continuing. Implement data loading/validation, preprocessing, baseline model or algorithm, training/processing, evaluation, saving predictions and basic plots. Prefer official reference implementations when licensing permits, and record any deliberate deviations. Do not present pseudocode as runnable code, rewrite unrelated modules, or claim successful execution without logs.

SAVE: Working baseline plus code understanding and prompt-to-code log.

PROMPT 10 | Verify and reproduce the baseline

> I have implemented [BASELINE] based on [PUBLISHED PAPER / OFFICIAL CODE]. Here are the code, dataset partition, configurations, actual logs and measured results: [FILES]. Audit the implementation against the original methodology: preprocessing, architecture, hyperparameters, random seeds, evaluation metric, data split and benchmark protocol. Design unit/sanity tests, inspect any discrepancy with reported results, and distinguish acceptable adaptations from changes that invalidate comparison. Produce a baseline validation report with VERIFIED, UNCERTAIN and NEEDS CORRECTION items, plus the exact additional checks required. Do not state that code is correct simply because it runs, and do not demand exact numerical replication under different conditions. Only after validation should we test extensions or comparisons.

SAVE: Baseline reproduction/validation report, including discrepancies.

PROMPT 11 | Debug using actual evidence

> My code failed or behaved unexpectedly. Relevant code: [CODE]. Actual traceback or observed output: [ACTUAL OUTPUT]. Expected behavior: [EXPECTED BEHAVIOR]. Determine the likely causes, separate confirmed evidence from hypotheses, and suggest the smallest safe correction with an explanation of each change. Provide a verification test and one relevant edge case; state whether earlier results may need to be rerun. Ask for more diagnostic information when needed. Do not claim to have run the code or generate imagined logs. Help me add this case to my AI-assistance/debugging log.

SAVE: Corrected code, tested fix and documented debugging decision.

PHASE IV — CONTRIBUTION AND EVIDENCE

PROMPT 12 | Choose and evaluate a defensible contribution

> My baseline [NAME] has been validated, and its actual results are [RESULTS]. My research question is [QUESTION], supported by [LITERATURE EVIDENCE]. Compare three feasible study pathways: A. reproduce an existing approach under a justified new condition; B. conduct a controlled comparative study of performance, generalization, robustness or efficiency; C. make a targeted, theoretically motivated extension or modification. For each, explain the scientific question, expected code changes, compute burden, advantages, failure risks and evidence needed. Recommend the most defensible pathway for my resources. Develop a fair test plan with appropriate comparisons and an ablation or failure analysis. If modification is chosen, guide me in making one isolated change while preserving the baseline implementation. Do not guarantee gains or mistake a routine comparison for novelty.

SAVE: Chosen contribution pathway and executable comparison plan.

PROMPT 13 | Evaluate using real outputs only

I have real experimental outputs in [CSV/LOG/FILE] for [METHODS], evaluated on [DATASET/SPLIT]. Write evaluation code that checks ID/label alignment, missing values, partitions and metric direction before computing [METRICS]. Compare methods under consistent conditions, summarize multiple runs or uncertainty where applicable, and export traceable results.csv plus publication-quality PNG/PDF figures. Show how to audit for leakage and incompatible comparator settings. Never invent example values inside the final evaluation tables or claim statistical significance without a valid test and assumptions. Ask for required files before interpreting my numerical results.

SAVE: Metrics script, actual result CSV, tables and research figures.

PROMPT 14 | Interpret outcomes, including negative results

My research questions are [QUESTIONS], protocol is [SETUP] and actual results are [TABLES/LOGS/FIGURES]. Analyze each question by separating direct observations, possible mechanisms, alternative explanations and unsupported claims. Discuss relative performance, variation across runs, computational cost, error or failure modes and limitations of the dataset/protocol. Explain which hypotheses are supported, contradicted or still unresolved. Recommend no more than three high-value follow-up experiments ranked by cost and likely information gained. Avoid exaggerating small effects, turning negative findings into positive claims or assuming a metric measures things outside its scope.

SAVE: Evidence-based Results and Discussion outline with limitations.

PHASE V — WRITE, REVIEW AND PUBLISH

PROMPT 15 | Draft an evidence-grounded manuscript in LaTeX

Use only my VERIFIED problem brief [BRIEF], original references [SOURCES], actual methodology [METHOD], experimental settings [SETUP] and measured results [RESULTS]. Help me prepare an IEEE/ACM-style paper in Overleaf. First create a section outline and evidence checklist, then draft Related Work, Methodology, Experimental Setup, Results and Discussion, Limitations, Introduction and Conclusion, and write the Abstract last so it matches the final evidence. Mark missing facts [TO VERIFY]. Preserve distinctions between previously published methods, my experimental contribution and conjectures. Create BibTeX entries only from metadata I have verified. Flag inconsistencies between figures, tables, code and claims, and help me resolve LaTeX compilation errors.

SAVE: 6–10-page target manuscript, figures, .tex and verified .bib.

PROMPT 16 | Conduct a demanding technical peer review

Review my draft manuscript [PDF] as a critical technical reviewer. Assess significance, literature coverage, contribution validity, technical correctness, data leakage, fair baselines, experimental design, robustness of conclusions, reproducibility, figure/table accuracy, citation traceability and clarity. Separate MAJOR and MINOR concerns, give section/page locations, explain why each concern matters and propose one concrete remedy or experiment. Identify unsupported novelty and performance claims. Do not default to praise, invent missing literature or treat a fluent manuscript as scientifically sound. Return a prioritized revision checklist and an author-response table.

SAVE: Peer-review report and manuscript revision log.

PROMPT 17 | Prepare reproducible GitHub documentation

Using my actual project files [FILES], environment [ENVIRONMENT], dataset access/license [RULES] and commands I have genuinely executed [COMMANDS], write a clear README with problem statement,

setup, dataset preparation, baseline and experimental run commands, evaluation, reproduction of key figures, compute requirements, limitations, credits, license and citation. Mark unexecuted commands as [TEST REQUIRED] rather than pretending they work. Check that configuration files and dependency versions are included, that paths are realistic, and that no secrets, private data or restricted model weights are exposed. Add a minimum reproduction checklist for a second student.

SAVE: Public-ready research repository and reproduction instructions.

PHASE VI — HUMAN–AI ACCOUNTABILITY

PROMPT 18 | Audit the human–AI collaboration

Using my authentic prompt log [PROMPTS], code commits/diffs [CHANGES], errors and debugging notes [NOTES] and experimental evidence [FILES], prepare a concise AI-use and research-integrity report. For each significant AI-assisted activity record: task, tool, prompt/version, AI proposal, human decision, verification method, code/result change and outcome. Provide at least three REAL examples of AI limitations, manual corrections or purposeful verification if available; if fewer occurred, report that honestly. Identify which conclusions depended on human reasoning, what was learned about vibe coding and which risks remain. Do not invent experiences, execution logs, fixes or performance gains.

SAVE: AI provenance report and reflective account of verified human decisions.

7 | Minimum experimental evidence package

- Traceable link from your previous review to a verified gap and a narrowly defined experimental question.
- Legally accessible dataset or measurement source with appropriate data handling and evaluation protocol.
- Executable baseline implementation plus a documented validation/reproduction check.
- At least two relevant methods/controls where appropriate, or a justified equivalent design.
- Predefined metric(s), consistent comparison conditions and saved results generated by actual code.
- At least one deeper evaluation: ablation, robustness, sensitivity, error/failure analysis or efficiency.
- Methodological limitations, negative findings and unresolved issues reported without concealment.
- Code and instructions sufficient to rerun the central experiment, subject to resource constraints.

Non-ML work is welcome. If your study concerns databases, systems, HCI, security or software engineering, choose rigorous domain-appropriate measurements rather than forcing an artificial accuracy metric.

8 | What to submit

Item	Required submission
A. Research paper	Final PDF, LaTeX source, .bib and figures. Recommended length: 6–10 pages excluding references; use instructor-selected template if specified.
B. GitHub repository	Source code/notebooks, README, dependency versions, configs, reproducible commands, dataset acquisition instructions and evaluation scripts.
C. Experimental evidence	Actual CSV/log files, evaluation outputs, comparison tables, figures, random seeds and documented test conditions.

Item	Required submission
D. Research provenance	One-page gap/decision brief, baseline validation report and concise AI prompt/code/verification log.
E. Demonstration	Five-minute code-and-results walkthrough plus readiness for a short technical viva or rerun of one component.

Suggested folder name: AITR_Lab02_EnrollmentNo_TopicShortName (or approved group identifier). Submission deadline, mode and group policy: as announced in Google Classroom.

9 | Repository structure

SUGGESTED PROJECT TREE

```
AITR_Lab02_Project/
  README.md              # research question and reproducibility
  requirements.txt       # verified dependencies
  configs/               # frozen experimental settings
  data/README.md         # access/description, not restricted raw data
  notebooks/             # exploration and verified runnable notebooks
  src/                   # reusable functions/models
  scripts/               # training, inference and evaluation entrypoints
  results/               # genuine metric files, logs and figures
  paper/                 # LaTeX source, bibliography and final figures
  research_decisions.md  # literature-to-experiment decisions
  baseline_validation.md # reproduction checks and deviations
  ai_assistance_log.md   # significant prompts and human verification
```

10 | Responsible AI and research-integrity rules

- Disclose AI support for ideation, literature synthesis, code generation, debugging, analysis, visualization and writing. Follow applicable institute and venue rules.
- Verify every important source, dataset, code/weight access claim and citation from its primary source. An AI-generated link is not proof.
- Understand, test and own the code you submit. Be ready to explain major modules, data flow and experimental choices.
- Never fabricate results, p-values, figures, execution logs, bibliographic entries, peer-review comments or debugging experiences.
- Avoid test-set tuning, data leakage, cherry-picking favorable runs and comparing methods under undisclosed unequal conditions.
- Respect licenses, dataset terms, research ethics, privacy and participant protections; never commit API keys or restricted data.
- Interpret similarity checks responsibly: similarity scores alone do not determine plagiarism or originality. Cite reused text, ideas and code appropriately.
- Report limitations and compute constraints openly; claims must remain proportional to the actual evidence.

11 | One-page research brief template

Use this template for Gate G1. Keep the response concise and evidence based.

Field	Complete with your project information
Review topic and key papers	[Topic]; 3–5 original references verified from the review
Research gap	[Specific limitation/unanswered question + source evidence]
Research question	[One primary question; optional supporting subquestions]
Study pathway	[Reproduction + extension / comparative / modification / hybrid]
Dataset or measurement setup	[Official source, access/license, planned partitions or protocol]
Baseline/comparators	[Verified method and availability of code/models]
Primary metric and control	[What will be measured and held constant]
Computing and risks	[Hardware, time, estimated bottlenecks and fallback]
First executable milestone	[Smallest piece of code/data verification completed or planned]

12 | Student final self-check

- ☐ My new research question follows directly from my earlier review, with original source evidence.
- ☐ The claims of gap and novelty are verified or carefully qualified.
- ☐ My dataset, license and evaluation protocol are recorded.
- ☐ My baseline runs, has been checked for correctness and is preserved for comparison.
- ☐ I know why my experimental design answers my stated research question.
- ☐ Every figure, metric and result is traceable to real code output.
- ☐ I conducted at least one deeper experiment or error analysis and reported limitations.
- ☐ My LaTeX manuscript distinguishes prior work, my changes and experimentally supported findings.
- ☐ My repository contains genuine commands, dependencies, configurations and no exposed secrets.
- ☐ My prompt log and technical decisions are authentic; I can explain and demonstrate the work.

THE STANDARD TO AIM FOR

Do not ask AI merely to write your next paper. Use AI to help you find a question, reproduce a reliable baseline, challenge assumptions, build a working study and generate evidence you can defend. Your code must run. Your results must be real. Your reasoning must be yours.